

Technical Specification: Bayesian Multilevel Regression and Post-stratification with Raking

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1 Overview

This document provides a technical specification of the Bayesian multilevel regression and post-stratification (MRP) model with raking used for population-aligned outcome estimation in trial–real-world comparisons. The model combines hierarchical outcome regression with probabilistic population alignment under marginal constraints.

Model development follows the Bayesian workflow framework, including prior predictive checks, prior testing using synthetic data, posterior predictive checks, and sensitivity analyses. Modeling decisions are guided by these diagnostics rather than fixed a priori choices.

2 Notation

Vectors are denoted by overbars (e.g. $\bar{p}$) or indexed definitions (e.g. $(\theta_c)_{c \in C}$), and matrices by bold uppercase letters (e.g. $\mathbf{L}$).

- $c \in C$: contingency table cells
- $k \in K$: baseline variables
- $j \in J_k$: levels of variable k
- (k, j) : variable–level combination
- $i \in \{1, \dots, T_{\text{original}}\}$: individual cases

3 Model Components

The Bayesian MRP with raking model consists of two linked components:

1. a multilevel outcome model
2. a raking model for population alignment

These components are connected through post-stratification.

3.1 Outcome Model

The outcome model is specified as a hierarchical ANOVA-style logistic regression with additive effects.

$$y_i \sim \text{Bern}(\theta_i)$$

$$\text{logit}(\theta_i) = \beta_0 + \sum_{k \in K} \beta_{k,j(i)}$$

The intercept follows $\beta_0 \sim \text{Normal}(0, 1.5)$. Level-specific coefficients follow Student- t priors:

$$\beta_{k,j} \sim \text{StudentT}(0, 4, \tau_k)$$

Scale parameters follow:

$$\tau_k \sim \text{HalfNormal}(\sigma_\tau), \quad \sigma_\tau \sim \text{HalfNormal}(0.7)$$

These priors provide weak regularization and enable partial pooling across levels while limiting over-shrinkage in moderately sparse data settings.

3.2 Raking Component

The raking component constructs a population-aligned contingency distribution that matches target marginals.

3.2.1 Deviation Model

The population-aligned cell probabilities are:

$$p_c = \frac{\exp(\eta_c)}{\sum_{c' \in C} \exp(\eta_{c'})}$$

with linear predictor:

$$\eta_c = \log(q_c) + \sum_{k \in K} \sigma_\delta \cdot \delta_{k,j(c)}$$

Deviation effects follow:

$$\delta_{k,j} \sim \text{Normal}(0, 1), \quad \sigma_\delta \sim \text{HalfNormal}(0.3)$$

This formulation preserves $\bar{p}$ as a smooth deformation of the observed distribution $\bar{q}$.

3.2.2 Marginal Alignment

The normalized contingency vector is scaled:

$$\bar{N} = T_{\text{target}} \bar{p}$$

Marginal counts are obtained via loading matrix $\mathbf{L}$ and aligned to target marginals:

$$\bar{N}_{\cdot\cdot}^* \sim \text{Poisson}(\mathbf{L} \bar{N})$$

Rows corresponding to zero target marginals are excluded from raking, as they receive zero weight in post-stratification.

3.3 Post-stratification

Cell-level outcome rates are:

$$\theta_c = \text{logit}^{-1} \left(\beta_0 + \sum_{k \in K} \beta_{k,j(c)} \right)$$

The population-aligned outcome rate is:

$$\theta_{\text{RWD} \rightarrow \text{RCT}} = \sum_{c \in C} p_c \theta_c$$

4 Sensitivity Analysis

Sensitivity focuses on prior scale parameters σ_τ and σ_δ . Posterior draws from the baseline model are reweighted via self-normalized importance sampling to reflect alternative prior specifications.

5 Graphical Model

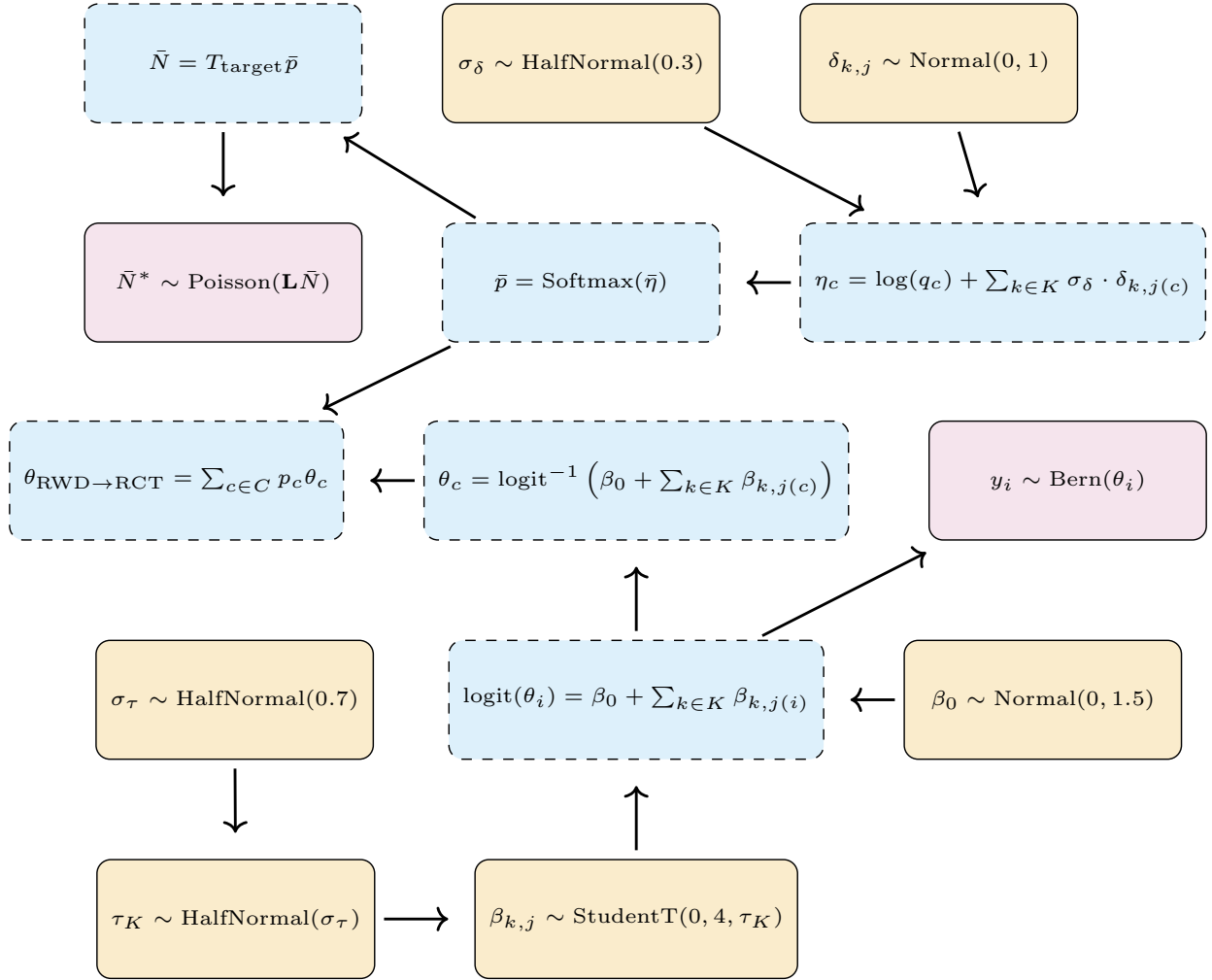


Figure 1: Bayesian MRP with raking. Blue boxes denote deterministic functions, orange boxes denote prior distributions, and purple boxes denote likelihood components. Arrows indicate conditional dependence in the generative model.